MILITARY SPECIFICATION
CENTER, MESSAGE AN/GSQ-80( )

1. SCOPE

1.1 Scope. This specification covers the modification of the shelter S-280/G and the installation of the applicable equipment and ancillary items into the modified shelter to form Message Center AN/GSQ-80( ).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

W-C-375	Circuit Breaker, Molded Case; Branch-circuit and service

MILITARY

MIL-P-116	Preservation, Methods of
MIL-W-6858	Welding, Resistance, Aluminum, Magnesium, Non Hardening steels or alloys, Nickel Alloys, Heat-Resisting Alloys, and Titanium Alloys, Spot and Seam
MIL-S-7124	Sealing-compound, Pressure-Cabin
MIL-T-7928	Terminals, Lug and Splice, Crimp Style, Copper
MIL-M-13231	Marking of Electronic Items
MIL-M-13787	Motors, Alternating current, Fractional Horsepower, Squirrel Cage
MIL-F-14072	Finishes for Ground Signal Equipment

FSC 9999

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MIL-S-552[illegible]	Shelter, Electrical Equipment S-28[illegible]/G

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for shipment and storage
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-252	Wired equipment, Classification of visual and mechanical defects

DRAWINGS

ELECTRONICS COMMAND DRAWINGS

SC-DL-51[illegible]	Message Center AN/GSQ-80( )

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AIR MOVING AND CONDITIONING ASSOCIATION INC.

Bulletin 210	AMCA Standard Test Code

(Application for copies should be addressed to the Air Moving and Conditioning Association Inc., 2159 Guardian Building, Detroit 26, Michigan)

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3. REQUIREMENTS

3.1 Description.- Message Center AN/GSQ-80( ), hereinafter referred to as message center, is a mobile assemblage, housed in a modified S-280( )/G shelter, with facilities for the receiving, sorting, distribution, and temporary storage of mail, messages, and dispatches. Message Center AN/GSQ-80 is capable of being transported by railroad, airlift, or 2½ ton truck.

3.2 Preproduction samples.- The contractor shall furnish one (1) preproduction sample of Message Center AN/GSQ-80( ) for approval unless otherwise specified by the invitation for bids and contract. (see 4.3).

3.3 Construction.-

3.3.1 Modified S-280( )/G Shelter.- Unless otherwise specified the Shelter, Electrical Equipment S-280( )/G shall be Government furnished. The S-280( )/G shelter shall be modified in accordance with the applicable drawings listed on SC-DL-512000.

3.3.2 Message Center Installation.- The Message Center AN/GSQ-80( ) shall be assembled by installing the equipment and items listed on SC-DL-512000 into the modified S-280/G shelter.

3.4 Vapor seal.- All joints of the exhaust ports, message drop, and power and signal entrance box shall be sealed to provide a vapor barrier against the entrance of moisture to the core material and to the interior of the shelter. All joints and edges shall be assembled and sealed in a manner to prevent collection and retention of moisture. All exterior rivets, screws, and fasteners shall be dipped in sealer prior to installation.

3.4.1 Sealer.- The sealer used as a vapor seal shall have the following properties:

(a) The sealer shall be a two part material consisting of an uncured compound and a separate catalyst in accordance with Specification MIL-S-7124.

(b) The minimum shear strength of the fully cured sealer and its bond to the aluminum shall be as follows:

at -65°F	at ambient temperature	at 200°F
200 psi	200 psi	75 psi

(c) The sealer shall withstand exposure to 100 percent relative humidity at 200°F for 2 weeks and have a shear strength of 75 psi minimum at 200°F.

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(a) After exposure to salt spray, 2% NaCl solution at 95°F for 2 weeks, the sealer shall have a shear strength of 200 psi minimum.

3.5 Cleaning.-

3.5.1 Parts.- After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable portion of Spec MIL-F-14072. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are assembled.

3.5.2 Units.- After assembly, the message center shall be cleaned thoroughly and shall be free from excess solder or flux, loose hardware, and other foreign material.

3.6 Finish.- The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings. (See 4.4)

3.7 Marking.-

3.7.1 General.- Marking shall conform to specification MIL-M-13231 and the equipment drawings. (See 4.4)

3.7.2 Nameplate.- Each message center shall be serial numbered and provided with a nameplate in accordance with the drawings.

3.7.3 Aircraft loading data plate.- The data plate for the S-280/G shelter shall be replaced with a data plate giving the weight and center of gravity of the message center and shall be in accordance with the drawings.

3.7.4 Interior and Exterior marking.- Interior and exterior markings shall be as depicted on the drawings.

3.8 Riveting.- Riveted joints shall be tight, the joined parts shall bee undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.9 Soldering.-

3.9.1 Solder.- Solder and flux shall be as defined on the drawings.

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3.9.2 Acid or acid salts.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections but no acid or acid salts shall be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.9.3 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.10 Tropicalization of material.- Canvas and materials as indicated on the drawings shall be treated to be resistant to moisture and fungi as follows:

3.10.1 Treating materials.- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.10.2 Toxicity.- Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished item when used for the purpose intended.

3.10.3 Flexibility.- Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

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3.10.4 Statement of treatment.- The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (See 4.4)

3.11 Welding.- Welded joints shall be as defined by the drawings and shall be such that grinding on the finished weld will be unnecessary. Spot, stitch, and seam welds shall conform to Specification MIL-W-6858, Class B. All surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from scale, paint, grease and other foreign materials. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Inert-gas-shielded arc welding shall be used for welding of aluminum except where spot welds are used. Welded assemblies shall be cleaned to remove any scale, oxidation products, and excess flux. Any acid used in cleaning shall be completely neutralized and removed. (See 4.4)

3.12 Wiring.- Wiring shall be in accordance with the drawings and shall be neat and sturdy. Wires shall be as short as practical except that sufficient slack shall be provided to facilitate field repair. Wires shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges. Wires in a continuous run between two terminals shall not be soldered during the wiring operation.

3.13 Crimping of solderless terminal lugs.- Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements and tests of Specification MIL-T-7928. (See 4.15)

3.14 Screws, threaded devices, and related parts.- Screws, threaded devices, and related parts shall be as defined by the drawings. Thread-forming screws shall not be used.

3.15 Securing of parts.- Threaded parts shall be tightened to a suitable value so as to retain parts securely when subjected to specified service conditions and tests.

3.16 Motors, fractional horsepower.- Fractional horsepower motors shall conform to Specification MIL-M-13787.

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3.17 Exhaust blowers.- Both exhaust blowers shall be the squirrel cage (centrifugal fan) type. Each blower assembly, as installed, shall satisfy the following minimum performance at atmospheric conditions corresponding to 70°F and 29.92 inches Hg.

Static pressure (inches of water)	.25	.50	.75	1.00	1.25
C F M	850	800	700	600	500

The exhaust blower motor shall operate on 115 volts, 60 cycle AC at a nominal 1725 rpm and full load watts shall not exceed 500 watts. (See 4.14)

3.18 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts, such as mounts, brackets, cabinets, etc., shall be physically and functionally interchangeable, without modification of such items, or of the shelter, or of the mounting locations. (See 4.17) Individual items shall not be hand-picked for performance or to fit the assigned position, mount or bracket. Reliance shall not be placed on an unspecified dimension, rating, characteristics, etc.

3.19 Military Truck transport.- The message center shall be capable of being transported over cross country terrain by military vehicle and shall withstand the test of 4.6. After testing per 4.6, there shall be no deformation, buckling, delamination, or damage to any part of the message center.

3.20 Rail Transportation.- The message center shall be capable of being loaded, blocked and braced on a flatcar for shipment by rail and shall withstand the test of 4.7. After each impact, there shall be no deformation, buckling, delamination or damage to any part of the message center.

3.21 Drops, flat and rotational.- The message center shall withstand drops of 18 inches onto concrete. After each of the drop tests of 4.8.1 and 4.8.2, there shall be no deformation, buckling, delamination, or damage to any part of the message center.

3.22 Watertightness.- The message center shall withstand the test of 4.9. After the test, there shall be no leakage into the message center or its walls, floor, roof, or doors.

3.23 Fording.- The message center, prepared for transportation on a military cargo truck, shall withstand the test of 4.10. After the test, there shall be no leakage into the message center or its walls, floor, or doors.

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3.24 Bounce.- The message center shall be tested on a vibration tester, as made by L.A.B. Corp. Type 10,000-SVMCT-12 or equal, modified for the required speed of 450 rpm with a 4" double amplitude. Table motion shall be circular-synchronous. Duration of test shall be one hour. Message center shall be tied-down and braced similar to Figure 1 with an additional vertical tiedown at each corner. After the test the message center shall be inspected and there shall be no damage or loosening of parts, screws, brackets, etc. (See 4.12)

3.25 Mounting Inserts.- All mounting inserts shall be installed without the use of any auxilliary materials such as adhesive and when installed the head shall not protrude more than 0.062 inch from the inside skin of the shelter. When installed each insert shall be capable of withstanding the tensile load and torque load referenced in para 4.16. There shall be no damage to the insert threads, the insert shall not have rotated from its original position, and there shall be no distortion of the panel, mounting member, or insert.

3.26 Power and Signal wiring.- Power and signal wiring shall be subjected to and pass the tests described in 4.11.

3.27 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs:

3.3 Construction
3.4 Vapor seal
3.5 Cleaning
3.6 Finish
3.7 Marking
3.8 Riveting
3.9 Soldering
3.10 Tropicalization of material
3.11 Welding
3.12 Wiring
3.13 Terminal lugs
3.15 Securing of parts

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be as classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3.)

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(b) Inspection covered by subsidiary documents. (See 4.4)

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.19)

4.3 Preproduction inspection.-This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table 1, the inspection specified in the subsidiary documents covering the items listed in 4.4 and the inspection specified for group A, group B, and group C (see tables II, III, and IV, respectively). The preproduction inspection shall be performed in this order.

(a) Military Truck, (b) Rail transport, (c) Flat drop, (d) Rotational drop, (e) Watertightness, (f) Fording, (g) Electrical, (h) Bounce. The remaining preproduction tests shall be performed in any order satisfactory to the contracting officer.

Table 1 - Preproduction inspection

Inspection (For additional preproduction inspection see 4.3)	Req't Para	Insp Para
Military Truck Transport	3.19	4.6
Rail transport	3.20	4.7
Flat drop	3.21	4.8.1
Rotational drop	3.21	4.8.2
Sealer	3.4.1	4.13
Exhaust blower	3.17	4.14
Bounce	3.24	4.12
Mounting Inserts	3.25	4.16

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4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery.

Item	Where Inspected
Finish	3.6
Marking	3.7
Welding	3.11

4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government may perform all or any part of the specified inspection, to verify that the item is in compliance with specified requirements. (See 4.6) Test equipment for measurement verification [illegible] shall be made available by the contractor.

4.5.1 Group A inspection.- This inspection, including sampling, shall conform to Table II and the inspection procedures of Standard MIL-STD-105, using normal inspection levels. Group A inspection shall be performed in any order which is satisfactory to the Government.

Table II -- Group A inspection

Inspection	Requirement Para	Inspection Para	AQL Major	AQL Minor
Visual and Mechanical	3.27	4.1?	.5 dpnu**	5.0 dpnu
Electrical	3.2?	4.1.	1.0?*	

* All electrical defects are considered major

** DPHU - defects per hundred units

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4.5.2 Group B inspection.- This inspection, including sampling, shall conform to table III and to the procedures for small-sample inspection of Standard MIL-STD-105, using the special inspection levels. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

4.5.2.1 Group B sampling plans.- The group B sampling plans, for AQL's listed in table III, shall be as follows:

AQL	Inspection level for normal and tightened and reduced inspection
4.0%	S-4
6.5%	S-4

4.5.2.2 Order of inspection within group B.- Group B inspection shall be performed in the order listed in table III.

Table III.- Group B inspection.

Inspection	Req't Para	Insp Para	AQL (see 4.5.2.1)
Production insert test	3.25	4.16.2	4.0%
Watertightness	3.22	4.9)	6.5%
Fording	3.23	4.10)	for group
Interchangeability	3.18	4.17	4.0%

4.5.3 Group C inspection.- This inspection shall be as listed in table IV, and shall normally be performed on sample units that have been subjected to and met group A and group B inspection.

Table IV.-Group C inspection

Inspection	Req't Para	Insp Para
Solderless-terminal-lug connection	3.13	4.15

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4.5.3.1 Sampling for inspection of solderless-terminal-lug connections.- Five specimens of each combination of wire and terminal shall be prepared each week for inspection. A specimen is a 1 foot length of wire on which a terminal lug has been applied using the same tools and procedures used in crimping the terminal lugs in production.

4.5.3.2 Noncompliance.- If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer. (See 6.3)

4.5.3 Reinspection of conforming group B and group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed Group B or Group C inspection, or both, may be accepted on contract, provided they are resubjected to and pass Group A inspection after repair of all visible damage.

4.6 Military Truck Transport.- The Message Center shall be transported on a Government furnished M-35 truck (a) on any convenient hard surfaced road and subjected to 60 abrupt stops from 30 miles per hour and (b) over the specially prepared course described in paragraph 4.6.1 for 10 laps at 5 mph, 10 laps at 10 mph, 10 laps at 15 mph, and 10 laps at 25 mph. A lap is defined as traversing the course in one direction. The message center shall meet the requirements of 3.10 after completion of both the (a) and (b) portion of the above test.

4.6.1 Military truck transport course.- The course for the military truck transport test shall have twelve 4 inch by 8 inch timbers placed 25 feet apart on the 8 inch face and with the 4 inch face fully above the ground. The edges shall have a 1 inch by 1 inch bevel. The sixth and twelfth timbers shall be placed 45 degrees to the direction of travel, and all other timbers shall be placed perpendicular to the direction of travel. The timbers shall be long enough to span the vehicle passing over them and shall be anchored securely by any convenient means.

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heat sealed when heat sealable barrier material is used. FSC 1740

4.7 Rail Transport, Impact Test.-The message center shall be loaded on a railroad flatcar (test car) in accordance with Figure 1. The test shall be conducted on a flat stretch of track. A loaded car totaling 165,000 lbs. (impact car) and traveling at 9 mph (or equivalent impact energy conditions approved by the Government) shall be impacted against the test car coupled to two other empty cars. These cars shall be stationary with all brakes off. Four impacts shall be performed, two impacts into each end of the test car. The cabling and blocking holding the message center in place shall be tight at the start of each impact run. If the cable or blocking is torn loose or if the load on the impact car shifts, the run shall be repeated. A damage survey shall be made after each impact and the message center shall meet the requirements of 3.20.

4.8 Drop tests.-

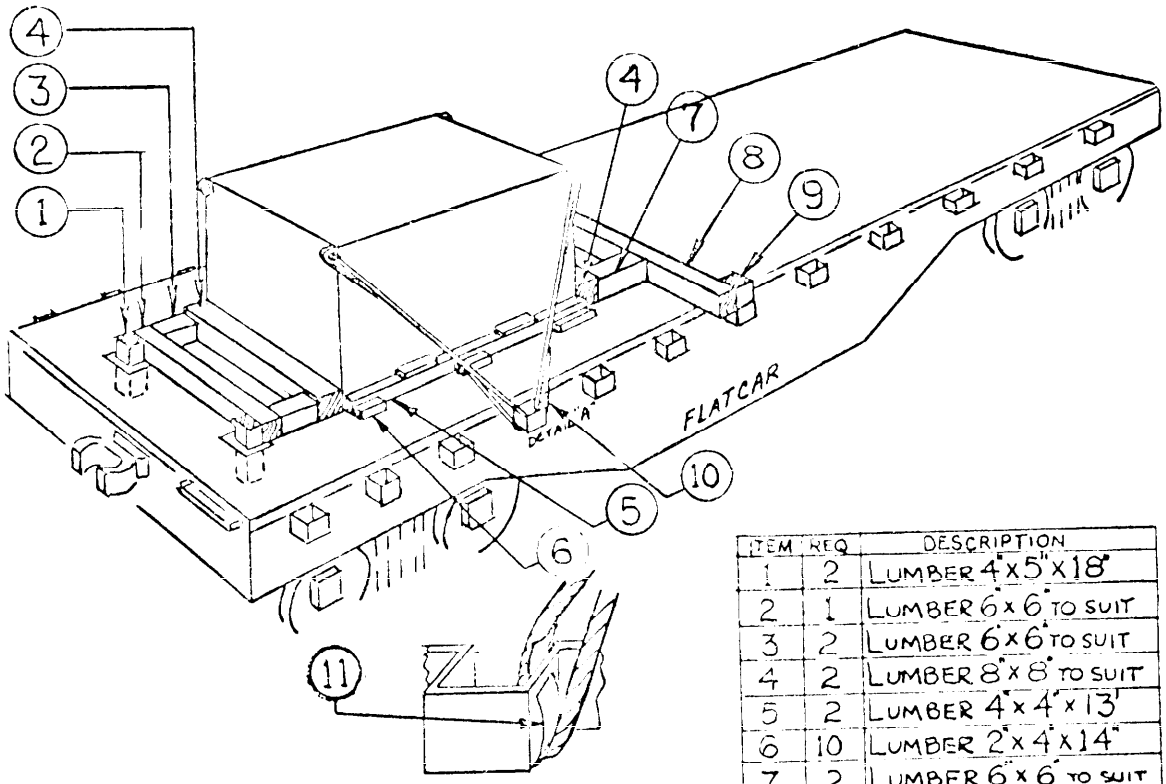
4.8.1 Flat drop.-The message center shall be lifted and allowed to fall freely a distance of 18 inches with the skids impacting onto a hard concrete surface. This test shall be performed once. Upon completion of this drop, the message center shall meet the requirements of 3.21.

4.8.2 Rotational drop.-The Message Center shall have one edge placed on a nominal 4 inch high board. The opposite edge of the message center shall be raised 18 inches and allowed to fall freely onto a hard concrete surface. This drop test shall be performed on each bottom edge of the message center. Upon completion of these drops, the message center shall meet the requirements of 3.21. In each drop test, the 4 inch high member shall be placed under the skids, and the eighteen inch measurement shall be taken from the bottom of the skids.

4.9 Watertightness test.-The message center, without the use of any additional caulking, sealing, taping, etc., shall be subjected to the following test: Each of the five exposed surfaces of the shelter, with covers closed, shall be sprayed with water from nozzles, model G29SQ (or CG29SQ) as made by Spraying Systems Co. Bellwood, Illinois, or equal, pointing directly at the surface and arranged to provide uniform coverage of the surface. Each end panel shall be tested using nine nozzles, each side using fifteen nozzles, and the roof using six nozzles. Each nozzle shall operate at a pressure of forty psig at the nozzle, and shall be located nineteen inches from the panel with the exception of the roof nozzles which shall be at a distance of thirty-two inches. (See Figure 2 for nozzle location.) Each panel shall be sprayed for a period of forty minutes. (If the watertightness test is to be followed by the fording test of 4.10, the bottom row of nozzles, with the exception of the nozzles facing the door-end panel, can be eliminated.) Upon completion of the above test, the message center shall meet the requirements of 3.22. To determine that no water has entered space between the skins, holes shall be made at points to be determined by the Government at time of testing. After this determination, the holes shall be sealed with rivets and sealer per 3.4.

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FIGURE 1



DETAIL "A"

Wire tiedown same for all points of contact with stake pockets

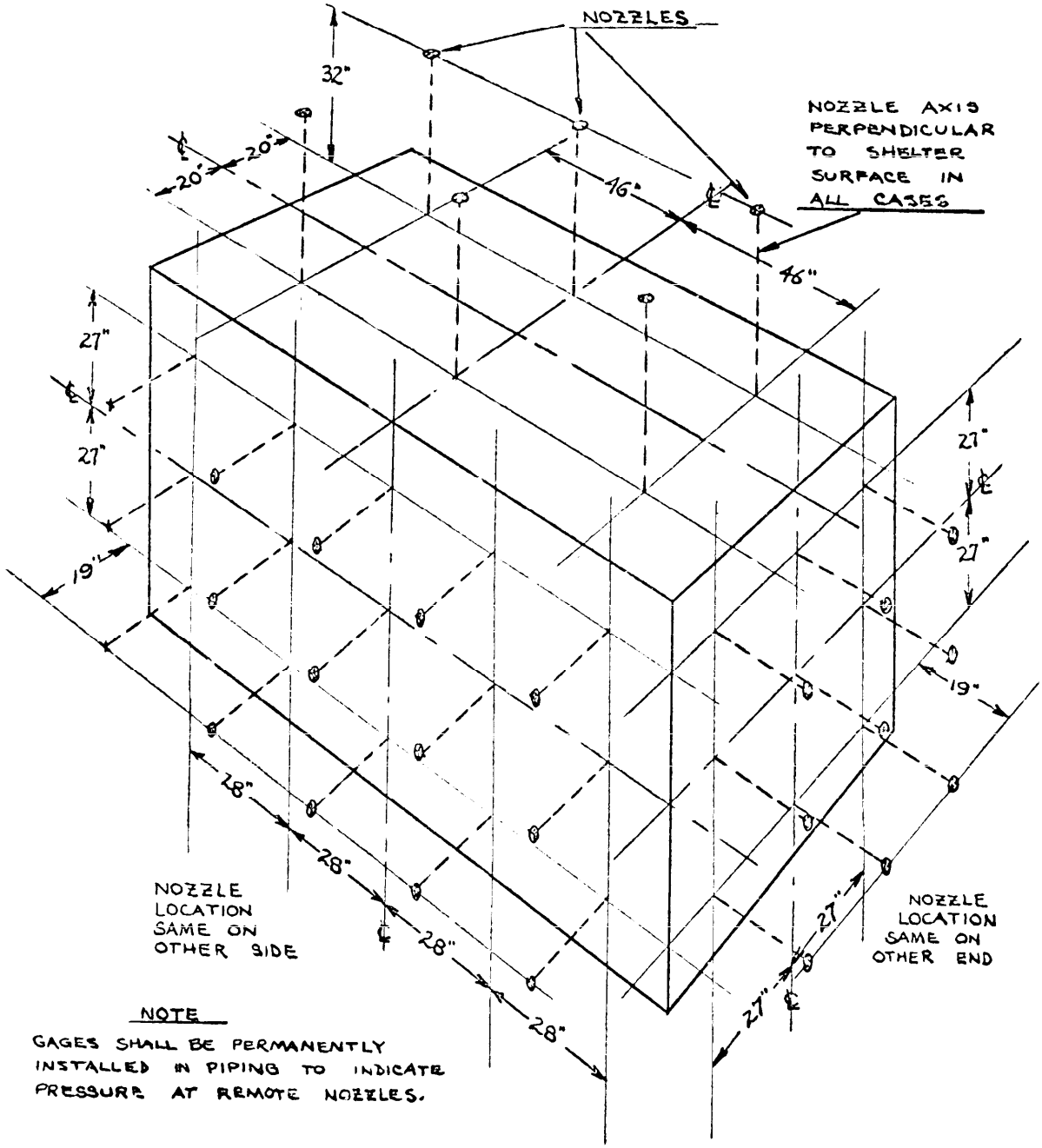
ITEM	REQ	DESCRIPTION
1	2	LUMBER 4"X5"X18"
2	1	LUMBER 6"X6" TO SUIT
3	2	LUMBER 6"X6" TO SUIT
4	2	LUMBER 8"X8" TO SUIT
5	2	LUMBER 4"X4"X13'
6	10	LUMBER 2"X4"X14"
7	2	LUMBER 6"X6" TO SUIT
8	1	LUMBER 6"X6" TO SUIT
9	2	LUMBER TO SUIT
10	4	WIRE #8 EXPORT (4 LOOPS)
11	4	FLEX. METAL SHEETING

NOTES:
a. The blocking lumber shall be notched to accomodate towing eyes, bolts and rivet heads.
b. The blocking lumber shall be rigidly fastened to the floor of the flatcar.

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FIGURE 2

NOZZLE LOCATION FOR TEST PER 4.9



NOTE

GAGES SHALL BE PERMANENTLY INSTALLED IN PIPING TO INDICATE PRESSURE AT REMOTE NOZZLES.

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4.10 Fording test.- The message center, prepared for transportation on a military cargo truck and without the use of any additional sealer, caulking, taping, fording kit, etc., shall be immersed in water to a depth of twenty-one inches, measured from the bottom of the shelter skids, for a period of two hours. The shelter shall be restrained by external tiedowns. Upon completion of this test, the message center shall meet the requirements of 3.23. To determine that no water has entered space between the skins, holes shall be made at points to be determined by the Government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer per 3.4.

4.11 Installation, electrical tests.- The following tests shall be performed on the electrical equipment in the Message Center prior to submission for acceptance.

4.11.1 Special test equipment.- In addition to other contractual requirements, special test equipment shall be provided by the contractor for preproduction and production testing of the AN/GSQ-60 configurations.

4.11.2 Insulation and continuity testing of power wiring circuits.-

(a) Remove all reactive and inductive loads from the line. Turn on all breakers and switches. [illegible] equipment, hi-pot between conductors and from each conductor to ground using a potential of 500 volts DC. The leakage resistance shall be a minimum of 1 megohm.

(b) Insert [illegible] convenience and equipment outlet individually. The [illegible] from the entrance receptacle to the outlet being tested shall be less than 5 ohms.

4.11.3 Power application.-

(a) Connect Message Center to a 115 volt ± 10%, 60 cycle ± 4%, single phase power source. Neon (power indicator) lamp shall be lighted.

(b) Switch on main breaker. Voltmeter shall read the input voltage.

(c) With auxiliary breaker in off position, the indicator lamp shall be out.

(d) Switch on each auxiliary breaker. Indicator lamp for each breaker shall be lighted.

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4.11.4 Test of lighting circuits.-

(a) Turn on blackout bypass switch and #1 light switch. Alternate ceiling lamps shall be lighted.

(b) Turn on #2 light switch. Remaining ceiling lamps shall be lighted.

(c) Turn off blackout bypass switch. Unlatch door. All lamps shall extinguisn before latch is in a full open position. Neon lamp shall remain lighted.

(d) Relatch door. All lamps shall be lighted when latch is in the locked position.

(e) Repeat (c) and (d) twice.

(f) Turn on blackout bypass switch and leave on for remainder of inspection. All lamps shall be lighted.

(g) Turn off circuit breaker #1. All lamps shall go out.

(h) Turn on circuit breaker #1. All lamps shall be lighted.

4.11.5 Test of exhaust blower circuits.-

(a) Plug in blower #1. Turn on switch for blower #1. Blower #1 shall operate. (Check that vent cover is open.)

(b) Turn off switch for blower #1. Blower #1 shall stop.

(c) Repeat (a) and (b) three times.

(d) Repeat (a), (b) and (c) for Blower #2.

(e) Turn on both blowers. Turn off circuit breaker #5. Both blowers shall stop.

(f) Turn on circuit breaker #5. Both blowers shall operate.

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4.11.6 Test of Electric Heater circuits.-

(a) Plug electric heater #1 into proper receptacle. Set switch on heater to "FAN" position. Heater shall blow air.

(b) Set switch on heater to "HEAT" position and turn thermostat to a temperature higher than ambient. Heater shall blow warm air after a few minutes of operation.

(c) Turn thermostat down to a temperature colder than ambient. Fan shall continue to blow air. Heating coil shall turn off indicated by a click and a drop in the ammeter reading.

(d) Set switch to a straight up position. Heater shall turn off.

(e) With heater switch in either operating position, turn off circuit breaker #3. Heater #1 shall shut off.

(f) Turn on circuit breaker #3. Heater shall operate again.

(g) Repeat tests (a) thru (f) for heater #2 and circuit breaker #4.

4.11.7 Test of intercom and equipment circuits.-

(a) Plug test lamp into each intercom and convenience outlet. Lamp shall light when plugged into each outlet.

(b) With test lamp plugged into any outlet, turn off circuit breaker #2. Lamp shall go out. Turn on circuit breaker #2. Test lamp shall relight.

4.11.8 Test of circuit breakers by ammeter.-

(a) With all equipment operating, including both electric heaters, the circuit breakers shall hold.

(b) Circuit breakers shall be tested in accordance with W-C-375. During this test, the ammeter shall indicate the load current.

4.11.9 Test of phone circuit.- Connect two wires from a test phone to binding posts 1A and 1B in signal entrance box. Connect tinned ends of a telephone cord to the binding posts on IA-312/PT mounted in Message Center, and insert plug end into phone jack. Operate ringer on each phone, in turn, and talk into each phone. Phones shall ring, and the conversation shall be loud and clear.

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4.11.10 Test of intercom circuit.- Connect a test intercom to binding posts 2A and 2B in signal entrance box. Plug power and signal connectors of [illegible] into appropriate receptacles. Turn on AC power to both intercoms. Operate the "OFF-SHED" switch to approximately midposition. Glow-lamp shall light. Operate "PRESS to TALK" switch of each intercom in turn, transmitting a test conversation in each direction. The conversation shall be loud and clear in both directions. Adjust the receiver and transmit volume controls to determine that they have the appropriate effect on volume. Operate the "OFF-SHED" switch to "off" position. Intercom shall still be able to receive a call.

4.12 Bounce test.- Message center shall be tested as required by paragraph 3.24. Test may be interrupted after each 5 minutes of testing for inspection. If corrections or adjustments are deemed necessary, the one hour of accumulated test time shall commence after the last correction or adjustment has been made. After completion of the test, the message center shall satisfy the requirements of 3.24. After completion of bounce test, the Message Center shall be subjected to the electrical tests of 4.11.

4.13 Sealer shear tests.- A tensile pull test shall be performed on a sample lap joints consisting of two one-inch wide strips of aluminum overlapped one-half-inch, giving contact area of one-half square inch and bonded with sealer material over no more than the specified one-half square inch area. The sample shall be tested under the conditions of 4.13.1 through 4.13.4. Five test specimens shall be used for each condition and tests shall be to destruction, with rate of application of load constant at one inch per minute maximum.

4.13.1 Ambient temperature shear.- Each sample shall be subjected to a tensile pull test at ambient temperature ($80^{\circ} \pm 15^{\circ}F$) and shall meet requirements of 3.4.1.

4.13.2 Temperature shear.- Five samples shall be tested at $-65^{\circ} \pm 5^{\circ}F$ and five shall be at $200^{\circ}F \pm 5^{\circ}F$. Samples shall meet requirements of 3.4.1. Temperature of sample shall be stabilized at specified level, as indicated by a thermocouple at the bond area.

4.13.3 Humidity exposure.- Each sample shall be tested after exposure to 100% relative humidity at $200^{\circ} \pm 5^{\circ}$ F for 2 weeks and shall meet requirements of 3.4.1. Temperature of samples shall be stabilized at specified level when tested.

4.13.4 Salt spray exposure.- Each sample shall be tested for conformance to 3.4.1 after being subjected to the salt spray test of Standard MIL-STD-202 using a 20% NaCl solution at $95^{\circ}F$ for a period of 2 weeks.

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4.14 Exhaust blower.- The exhaust blower shall be tested in accordance with the AMCA Standard Test Code Bulletin 210. In the motor rpm and wattage shall be recorded for each test point. The exhaust blower test results shall indicate compliance with 3.17.

4.15 Testing of solderless terminal - lug connections.- Connections made with solderless terminal - lugs shall be tested to meet the requirements of 3.13

4.16 Mounting Inserts.

4.16.1 Preproduction Insert test.- Four of each size insert mounted in opposite panels; eg, floor and ceiling or side and side shall be tested with the axial load, specified in Table V, applied to eye bolt in the insert. After axial load is removed the specified torque load (Table V) shall be applied to a bolt in the insert. (Note: Bolt shall lock on threads in insert and not "bottom" nor shall the bolt head be tight against head of insert.) After test, the requirements of 3.25 shall be met.

Table V.-Preproduction Insert Loads

Insert size	Tension (lbs)	Torque (in lbs)
5/16 - 18	2000	40
1/4 - 20	1300	23

4.16.2 Production Insert test.-Two percent of the total number of any size insert, but not less than five inserts, shall be randomly selected for test in each sample assemblage. Each selected inserts shall have a torque load, in accordance with Table VI applied to the bolt. (Note: Bolt shall hold a stationary plate against head of insert). After test, the requirements of 3.25 shall be met. Failure of any insert shall be cause for rejection of sample assemblage.

Table VI.-Production Torque Load

Insert size	Torque (in lbs)
5/16 - 18	100
1/4 - 20	60
10 - 32	30
8 - 32	20

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4.17 Inspection for Interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.13. When a dimension is not within specified or design limits, it shall be considered a defect.

a. Location and size of mounting inserts.
b. Size and mounting dimensions of brackets.
c. Size and mounting dimensions of drawers, cabinets, and racks.
d. Size and mounting dimensions of slide board and table tops.
e. Size capacity and mounting dimensions of blowers, and heaters.

4.18 Visual and mechanical inspection.- Equipment shall be examined for the defects listed in Standard MIL-STD-252 and table VII.

Table VII.-Classification of visual and mechanical defects

Classification	Defects
Major	1. Mounting insert missing or mislocated.
	2. Sealer improperly applied (holes, separations, or lack of adhesion).
	3. Welds - cracked or porous.
	4. Parts, brackets, or equipment loose or missing.
	5. Drawers, doors, etc. inoperative.
	6. Gaps greater than 1/16" between sections of wire duct.
	7. Locking or holding devices inoperative.
	8. Lock washer or lock nut missing.
	9. Loctite not used where directed by drawing.
	10. Incorrect or missing aircraft data loading plate.
Minor	1. Screw loose
	2. Drawers, doors, etc. difficult to operate.
	3. Locking or holding devices difficult to operate.
	4. Finish - abrasions or scratches.

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4.19 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of section 5.

4.20 Rough handling test (preparation for delivery).- When rough handling test in accordance with Specification MIL-P-116 is required by the contract (see 6.1), the following functional tests shall be conducted to determine freedom from operational malfunction caused by the rough handling:

Paragraph 4.18 Visual and mechanical.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging.- AN/GSQ-80 shall be packaged in accordance with Specification MIL-S-55286, for the levels as specified. (See Section 6).

5.2 Packing.- AN/GSQ-80 shall be packed in accordance with Specification MIL-S-55286, for the levels as specified. (See section 6).

5.3 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with the applicable requirements of MIL-STD-129.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment (Level A, level B, or level C.)

(c) The specific paragraphs of section 5 which are applicable to particular procurement.

(d) Preproduction pack(s) as follows:

(1) Makeup of pack (s).
(2) Number of each kind of pack to be submitted.
(3) Inspection to be performed thereon--including rough handling test, which will not be performed as procurement inspection.
(4) Five copies of process sheet, using format shown on Drawing SC-C-33073 and accompanied by such drawings, sketches, and other data as may be necessary to completely describe the procedure followed in fabrication of the preproduction pack(s) and to identify the items therein. This data should be submitted to the contracting officer with the preproduction pack.
(5) A packaging technican from the procuring agency will evaluate the preproduction pack(s).

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(e) Marking and shipping of samples.
(f) Place of final inspection.
(g) List of each individual equipment item or equivalent dummy load, with details, to be used in testing the shelter.
(h) Submission of the statement of treatment referenced in 3.10.4, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.2 Definitions.-

6.2.1 Inspection.-Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.2.2 Examination.-Examination consists of simple, generally non-destructive determinations of compliance, without use of special testing equipment.

6.2.3 Testing.-Testing consists of determinations of compliance, using technical means.

6.3 Group C inspection.-Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2).

6.4 Nomenclature.-The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/GSQ-80(X). The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.5 Verification inspection.-Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian:
Army (EL)

Preparing Activity:
Army (EL)

Project Number 9999-A003

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119 R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of pr
s for ultimate use by the Department of Defense. This sheet is provided [illegible] obtaining information on the use of this specification which will
ure that suitable products can be procured with a minimum expenditure of time and effort at the lowest possible cost. Comments and the return of this form will be
preciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to prepering activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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